Shuai Hao

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EDUCATION

Columbia University New York, NY
Dual Degree M.S. in Computer Science and M.S. in Journalism 08/01/2018 – 05/20/2020

- Coursework: Computational Journalism, Data Visualization, Database, Natural Language Processing and User Interface Design

Communication University of China Beijing, CN
B.A. in Journalism, Minor: Computer Science (Ranked in top 5% of the class) 09/01/2014 – 07/01/2018

- Coursework: Data Structure, Web Design & Improvement, Probability & Statistics and Multimedia Reporting & Editing

TECHNICAL SKILLS

- **Programming:** C, C++, Python, R, Ruby, SQL, PHP, HTML, CSS (Tailwind), JavaScript (D3, React, Svelte, TypeScript)
- **Data Analysis:** Gephi, Google Colaboratory, Jupyter Notebook, KNIME, Observable, OpenRefine, QGIS, SPSS, Tableau
- **Digital:** Adobe Suite, Bloomberg Terminal, Datawrapper, Figma, Git, Infogram, Linux, Mapbox, Microsoft Suite, WordPress

PROFESSIONAL EXPERIENCE

Moody's Ratings New York, NY
Software Engineer 07/06/2026 – Now

- Built and maintained a charting library using D3, Three.js and Svelte, enabling the team to rapidly create on-brand visualizations
- Managed AWS S3 buckets, Lambda function, APIs, and data pipelines, enhancing data reliability and operational efficiency
- Leveraged GitHub Copilot and other generative AI tools to accelerate code generation, debugging, and code review

Associate Software Engineer 02/07/2022 – 07/05/2026

- Developed nine web applications that convert spreadsheet data into customizable SVG and PNG visualizations
- Built a web application with a zoomable collapsible tree, a sidebar and search/filter functions to show all research themes
- Set up two data story templates and an UI library to automate, templatize and accelerate the creation of data visualizations
- Produced mobile-first, interactive data stories by transforming complex credit research data into engaging narrative.

CoinDesk New York, NY
Data Visualization Analyst 06/08/2020 – 02/04/2022

- Scraped, wrangled and parsed financial data on Google Colab for CoinDesk Index products and quarterly reviews
- Built 17 automated chart templates in D3.js to visualize price, market capitalization, volatility and correlations of crypto and macro assets. Journalists can use templates to generate custom charts, embed them in the CMS and download data for analysis
- Created over 400 charts using Tableau, Datawrapper and Photoshop for daily news stories and social media posts
- Set up an interactive dashboard using SQL to visualize how Ethereum's London upgrade improved the blockchain

USA TODAY Network McLean, VA
Interactive Graphics Intern 06/18/2019 – 08/23/2019

- Developed five visual data projects on politics, education, history and health. Work included analyzing and cleaning data in Pandas, extracting pixel values of images in PIL and making D3 animations. Created interactive maps, bubble charts and SVG transformations for a node.js project about the history of slavery in the U.S., which received more than 300,000 pageviews
- Improved and updated a timeline template by reorganizing elements on the web page in Angular.js for a story about how women fought for suffrage. Used Adobe Illustrator and Photoshop to design and polish a map about marijuana legislation

Caixin Media Beijing, CN
Data Journalism Intern 06/19/2017 – 09/19/2017

- Analyzed large databases and visualized launching sites, maximum range and payload of missiles with an interactive chart in D3.js and two static charts in Adobe Illustrator to display increasing security threats from North Korean missiles
- Designed three interactive webpages examining topics such as school bus regulations in China and the U.S., geographic positions between Chinese universities and the rank of rising billionaires in China

PROJECTS

Web Development | Leverage Finance Data Product Apr. 2025 - Now

- Led development of a public-facing interactive dashboard processing 500K+ rows of data, featuring multi-level filtering across 15 dynamic charts. Built with D3.js, Svelte, and TypeScript, achieving low-latency performance through browser-side optimization and caching strategies. Maintained the platform with monthly data updates and financial analysis